

Serendipity Wall: A Discussion Support System Using Real-time Speech Recognition and Large Language Model

Shota Imamura*
The University of Tokyo

Hirotsuka Hiraki†
The University of Tokyo

Jun Rekimoto‡
The University of Tokyo
Sony CSL Kyoto

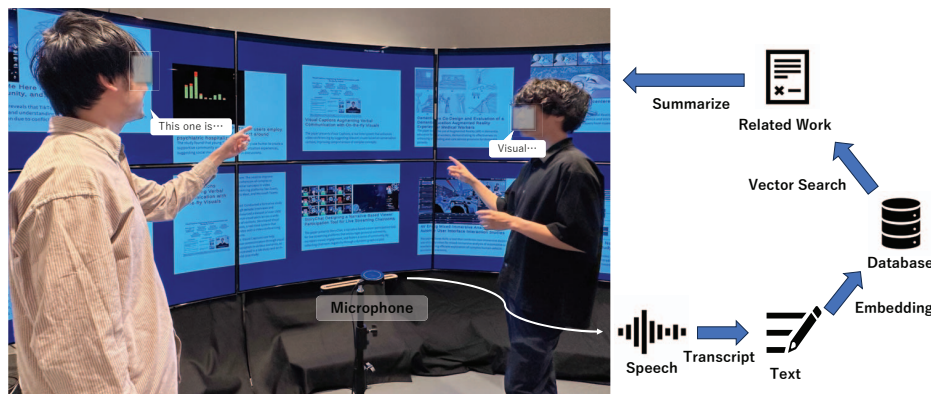


Figure 1: The Serendipity Wall is a system that facilitates discussion and new insights by presenting relevant information according to the content of the discussion. The system collects and transcribes the utterances by participants of the ongoing discussion in real time. The transcribed sentences are divided into chunks of time and are converted into embedded vectors. Then, vector searching extracts the information relevant to the discussion from the database. The extracted information is summarized by a large-scale language model and are presented on the large display.

ABSTRACT

Group discussions are important for exploring new ideas. One method to support discussions is presenting relevant keywords or images. However, the context of the conversation and information tended not to be taken into account. Therefore, we propose a system that develops group discussions by presenting related information in response to discussions. As a specific example, this study addressed academic discussions among HCI researchers. During brainstorming sessions, the system continuously transcribes the dialogue and generates embedding vectors of the discussions. These vectors are matched against those of existing research articles to identify relevant studies. Then, the system presented relevant studies on the large display with summarizing by an LLM. In a case study, this system had the effect of broadening the topics of discussion and facilitating the acquisition of new knowledge. A larger display area is desirable in terms of information volume and size. Therefore, in addition to large displays, virtual reality environments with headsets could be suitable for this system.

Index Terms: Human-centered computing—Human computer interaction (HCI);

1 INTRODUCTION

Brainstorming and group discussion are essential collaborative activities for creative problem solving and ideation, and are widely used in product development, research, marketing, and many other

areas [6]. Prior research has proposed tools to facilitate such idea generation and facilitate discussions [2].

Among them, a method to expand topics using Word2Vec [5] has been proposed as a method to support discussion based on its conversation [4], and a method to visually activate the discussion by presenting the words themselves or images associated with the web articles that have emerged from the discussion [8] [1] [7]. Methods to provide such ideation support in the context of a conversation in a VR space has also been proposed. Conversation Echo [3] supports inspiration by generating a VR space with 360-degree images in real-time based on conversations.

Based on these previous studies, it is expected that information presentation tailored to the content of the conversation will broaden people's ideas and cognitive abilities. However, prior studies have not provided discussion support based on individual documents. In addition, existing studies on conversations during discussions only provide word-based support, such as the presentation of synonyms in Word2Vec using words or the Wikipedia dataset, and do not provide discussion support based on the context of the conversation.

In this study, we propose Serendipity Wall (Figure 1), a discussion support system, that generates embedded vectors based on transcriptions of discussions and searches for related information from databases. It presents information that is close to the ongoing discussion, facilitating brainstorming and expanding knowledge.

2 IMPLICATION

2.1 equipment

The Serendipity Wall consists of output devices for information presentation, a microphone for sound collection, a front-end system for information presentation using Unity, a local FastAPI server for information retrieval, a database of related information, and a PC for processing the information.

*e-mail: imamura.shota.ap@alumni.tsukuba.ac.jp

†e-mail:hiraki-uts1@g.ecc.u-tokyo.ac.jp

‡e-mail:rekimoto@acm.org

Large displays, projectors, or multiple head-mounted displays can be considered as output devices so that multiple people can view various information presented during face-to-face discussions at the same time. The case study used a large display space consisting of six LG 65EG9600 curved 4K OLED displays combined as an integrated display. ReSpeaker USB Mic Array microphones from Sseed Studio were used as microphones.

2.2 Embedding vectors and Vector search

In this study, the embedding vectors were generated using the Embedding API provided by OpenAI. The system was implemented for the example use case of discussions of Human Computer Interaction research ideas. The relevant information to be presented was a set of papers from CHI2023. The first page of the papers contains the title, abstract, keywords, and caption information for the teaser image, which is expected to summarize the contents of each study. Therefore, we decided to use the first page of each research paper as the embedding vector in the implementation of this study. In addition, to facilitate reflecting the direction of the latest conversation, we decided to use the transcriptions of every 20 seconds of the content of the conversation as the embedding vector of a single conversation. To search for papers that were highly relevant to the content of the conversation, a vector search was performed using FAISS, a library that can perform efficient similarity searches of vectors.

The information was initially presented by entering the topic of the discussion and displaying the top 10 vector search results that came up, showing the top 10 relevant studies (Figure 2). Thereafter, the closest study to the conversation's embedding vector was added every 20 seconds, pushing out the least relevant related studies.



Figure 2: Use case image in a virtual space.

Two types of display format for each study were prepared with different amounts of information (Figure 3). One displayed the key visual, the title, and a short summary using a large-scale language model. The short summary gave the context of the first page of the paper and prompted the user to summarize the main points and arguments in 30 words or less. OpenAI's ChatGPT API (GPT-4) was used for the large-scale language model. Because a short summary alone does not provide enough information, another display format was also provided: a key visual, a title, and a summary of the issue, methodology, and results using the large-scale language model. The detailed information was presented when a study that had already been displayed came to the top of the list again in the vector search.

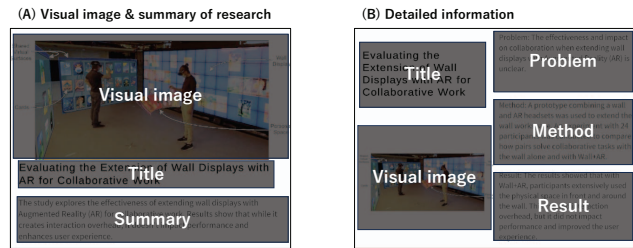


Figure 3: (A) Key visual image and title and summary are combined in one panel. (B) The more detailed information version consists of a title visual image and summaries of the problem, method, and result.

3 CASE STUDY

To evaluate the proposed method in supporting discussions about HCI research, we asked participants to engage in discussions about HCI research. 6 participants from HCI-related laboratories were recruited, making a total of 3 pairs of 2 participants each.

From participant's feedbacks and the expansion of the embedding vector of the conversation, we found that presenting a summary of research related to discussions triggered conversation, allowed participants to obtain related research and ideas, and expanded the discussions. The system facilitates discussion, such as moving the discussion from vague concepts to specifics based on relevant studies and communication through the use of pictures.

On the other hand, it was also found that the current system is implemented as presenting other related studies one after another, which has the disadvantage of diverging the conversation and making it difficult to deepen the discussion.

4 CONCLUSION

In this study, we proposed Serendipity Wall, a system that facilitates discussion by presenting information relevant to the discussion using LLM and vector search. In a case study, HCI research discussions were conducted using the Serendipity Wall, and we confirmed that the system has the effect of stimulating discussion and broadening the conversation by gaining new knowledge, facilitating discussion through information, and making ambiguous interests more concrete. On the other hand, it also became clear that depending on the way information is presented, there may be side effects that prevent the discussion from deepening.

In the future, it will be important for content generation to consider more detail attention based on gaze movements. We believe that the proposed system will be more useful by using VR headsets that can easily accommodate eye-trackers and widely distribute information throughout its virtual space.

In addition, this system retrieves information in cyberspace based on implicit interaction through conversation and composes a "space" for presenting that information. Therefore, the system is expected to have potential as an interaction method that seamlessly links physical space and cyberspace by composing a space based on attention inferred from conversation and gaze movements.

ACKNOWLEDGMENTS

This work was supported by JST Moonshot R&D Grant JPMJMS2012, JST CREST Grant JPMJCR17A3, and the commissioned research by NICT Japan Grant JPJ012368C02901, JST ASPIRE Program Grant Number JPMJAP2327.

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